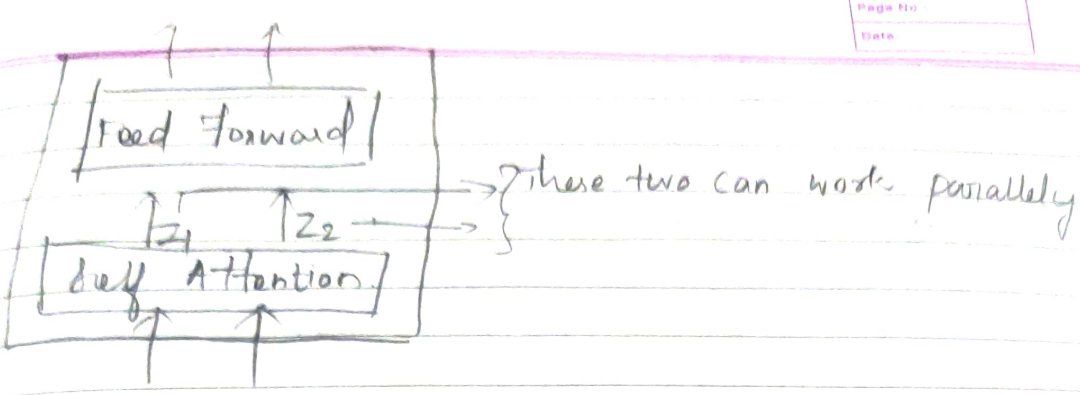


02/10/2026

Lecture 45,46

Positional Encoding:-

	S_1	Index	Embedding vector.
x_1	I	0	.. .. 512dim
x_2	love	1	.. .. 512dim
x_3	AI	2	.. .. 512dim

The come up method:-

S_1	Index (Or) position	Embedding vector	Positional Embedding vector	Input ⇒ Transformer
x_1 I	0	1X512	1X512	

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{2i/d}}\right)$$

$$PE(pos, 2i+1) = \cos\left(\frac{pos}{10000^{2i/d}}\right)$$

Where

d = embedding dimension of a token

i = index runs across positional encoding vector.

Token	word embedding vector	position	positional encoding vector	Final input vector
I	[1.0, 0, 0]	0	[ ]	[ ]
love	[0, 1, 0, 0]	1	[ ]	[ ]
AI	[0, 0, 1, 0]	2	[ ]	[ ]

d = 4

Alternate position

Even ⇒ sin functions.

odd ⇒ cos functions

Even ⇒ $PE(pos, 2i) = \sin\left(\frac{Pos}{10000^{2i/d}}\right)$

odd ⇒ $PE(pos, 2i+1) = \cos\left(\frac{Pos}{10000^{2i/d}}\right)$

Position 0 i - index / position

0	1	2	3
0	1	0	1

i = 0 $PE(pos, 2i)$ 2×0

sin

$\Rightarrow PE(0,0) = \sin\left(\frac{0}{10000^{0/4}}\right)$

$= \sin(0) = 0.$

i = 0 ~~pos / 2i~~

cos

$PE(pos, 2i+1) = \cos\left(\frac{0}{10000^{0/4}}\right)$

$PE(0,1) = \cos(0)$

$= 1.$

pos = 0

$PE(pos, 2i)$

$PE(0, 2) = \sin\left(\frac{0}{10000^{2/4}}\right)$

$= \sin\left(\frac{0}{10000^{1/2}}\right)$

$= \sin(0/100) = \sin(0)$

$= 0.$

pos = 0, i = 1

$PE(pos, 2i+1)$

$PE(0, 3) = \cos\left(\frac{0}{10000^{2/4}}\right)$

$= \cos(0) = 1$



Solution 1

Rad. DEG

i = 0

$PE(1, 0) = \sin(1/10000^{0/4})$

$= \sin(1/1) = \sin 1 = 0.8414$ 0.017

i = 0

$PE(1, 1) = \cos(1/10000^{1/2})$

$= \cos(1/100) = \cos(0.01) = 0.999$ 0.999

i = 1

$PE(1, 2) = \sin(1/10000^{1/2}) = \sin =$ 0.00017

i = 1

$PE(1, 3) = \cos(1/10000^{1/2}) = \cos(1/100) =$ 0.999